LSRIF: Logic-Structured Reinforcement Learning for Instruction Following

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Abstract

Instruction-following is critical for large language models, but real-world instructions often contain logical structures such as sequential dependencies and conditional branching. Existing methods typically construct datasets with parallel constraints and optimize average rewards, ignoring logical dependencies and yielding noisy signals. We propose a logic-structured training framework LSRIF that explicitly models instruction logic. We first construct a dataset LSRINSTRUCT with constraint structures such as parallel, sequential, and conditional types, and then design structure-aware rewarding method LSRM including average aggregation for parallel structures, failure-penalty propagation for sequential structures, and selective rewards for conditional branches. Experiments show LSRIF brings significant improvements in instruction-following (in-domain and out-of-domain) and general reasoning. Analysis reveals that learning with explicit logic structures brings parameter updates in attention layers and sharpens token-level attention to constraints and logical operators.

1 Introduction

Instruction following is a core capability of large language models (LLMs) and is essential for their use in real-world applications (Zhang et al., 2025; Lu et al., 2025; Ye et al., 2025). User instructions are often complex and may span multiple turns or agent-based interactions (Qi et al., 2025; Deshpande et al., 2025). Beyond producing fluent text, effective instruction following requires models to correctly understand and satisfy multiple constraints, which are often expressed through structured and interdependent conditions (He et al., 2024; An et al., 2025).

In essence, complex instructions are composed of multiple constraints connected by logical structures. Correct instruction following therefore requires not only satisfying individual constraints, but also adhering to the logical relationships between them. As shown in Fig. 1, the complex instruction contains three common types of logical relationships. Parallel (And) structures require all constraints to be satisfied simultaneously. Sequential (First–Then–Finally) structures impose an execution order, where later constraints depend on the successful completion of earlier ones. Conditional (If–Else) structures introduce branching logic, where the model must first evaluate a condition and then follow the correct branch.

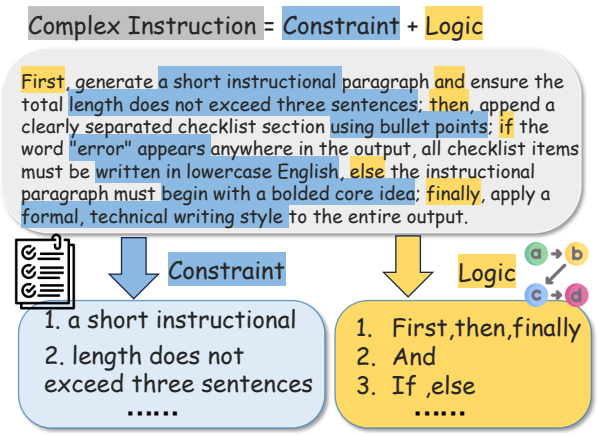

Figure 1: Essentially, the complex instruction is the logical composition of constraints.

Existing approaches for improving instruction following still face clear limitations when dealing with logically structured instructions. From the perspective of **data construction**, most training data simplify instructions by treating all constraints as parallel (Sun et al., 2024; Huang et al., 2025). Although some datasets include logical structure, they are mainly used for evaluation rather than training (Wen et al., 2024; Wang et al., 2025). In terms of **reward modeling**, the reward for the entire instruction is often computed as the average of the rewards for individual constraints (Qin et al.,

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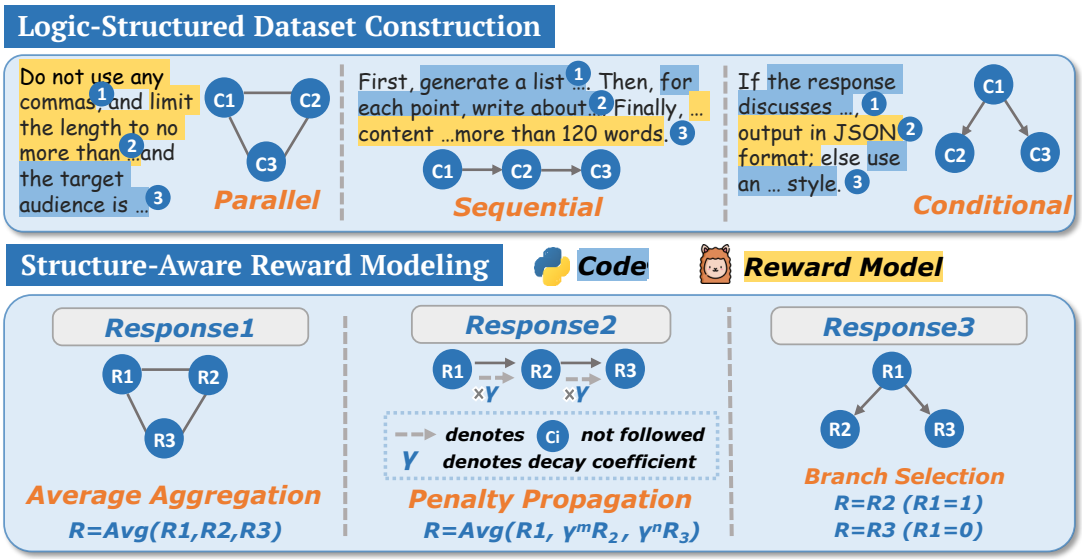

Figure 2: Our framework LSRIF consists of two components: (LSRINSTRUCT) logic-structured dataset construction, and (LSRM) structure-aware reward modeling with corresponding methods.

2025). This assumes that constraints are independent. However, for sequential or conditional instructions, failure at an early step makes later constraints irrelevant, and simple averaging can produce incorrect training signals. Finally, regarding **interpretability for performance improvements**, prior work typically shows gains in instruction-following performance and the preservation of general reasoning abilities (Peng et al., 2025), yet the underlying reasons remain unexplored. Furthermore, it remains unclear whether gains in logically structured instruction following actually transfer to reasoning ability.

To address these limitations, we propose a logic-structured training framework LSRIF that explicitly models instruction logic in both data construction and reward design. (1)**Logic-Structured Data (LSRINSTRUCT)**. We define instruction structures using three basic logical forms: parallel, sequential, and conditional. Based on these forms, we construct a dataset of multi-constraint instructions covering multiple logical structures. (2) **Logic-Structured Reward Modeling (LSRM)**. We design reward modeling methods that reflect the execution semantics of different structures. For parallel structures, rewards are aggregated by averaging. For sequential structures, we apply a decay mechanism so that failures in earlier steps reduce rewards for later ones. For conditional structures, rewards are assigned only to the constraints in the correct branch. (3) **Interpretability for Performance Improvements**. We further analyze how logic-structured training affects the model. We observe larger parameter updates in attention layers than in MLP layers. At the token level, trained models place more attention on logical connectors and constraint-related tokens. These changes also appear in general reasoning tasks, indicating that the learned ability transfers beyond instruction following.

Our contributions are summarized as follows: (1) We propose LSRIF, a logic-structured training framework. (2) LSRIF includes LSRINSTRUCT, an instruction dataset capturing parallel, sequential, and conditional constraint logic structures, and LSRM, structure-aware reward modeling that aligns reward signals with logical execution semantics. (3) LSRIF improves both in-domain and out-of-domain instruction-following performance and general reasoning ability, with attention and token-level interpretability analysis.

2 Related Work

2.1 Instruction Following Data Construction

Existing work constructs datasets with multi-constraint instructions to improve instruction-following capabilities (Qin et al., 2025; Cheng et al., 2024). However, these approaches directly concatenate constraints, ignoring potential structures among them, which fails to simulate real-world user instructions. While some datasets consider logical structures (Wen et al., 2024; Wang et al., 2025), they are primarily designed for evaluation rather than training. In contrast, we construct

a training dataset where constraints show explicit logical structures.

2.2 Reward Modeling for Instruction Following

Training paradigms for instruction following have evolved from supervised fine-tuning (Sun et al., 2024) to Direct Preference Optimization (Huang et al., 2025; Qi et al., 2024) and Reinforcement Learning with Verifiable Rewards (RLVR) (Peng et al., 2025; Qin et al., 2025). Existing RLVR methods aggregate constraint-level rewards through simple averaging. However, this averaging strategy fails when constraint logical structures are not parallel (e.g., sequential or conditional). We propose structure-aware reward modeling, where different structures employ distinct reward modeling methods.

3 Method

Our approach consists of two main components: logic-structured dataset construction (LSRINSTRUCT) and structure-aware reward modeling (LSRM). As illustrated in Fig. 2, we organize instructions into three logical structures—Parallel, Sequential, and Conditional and employ a structure-aware reward model with three corresponding methods: Average Aggregation for parallel structures, Penalty Propagation for sequential structures, and Branch Selection for conditional structures.

3.1 Logic-Structured Dataset Construction

To move beyond flat constraint concatenation, we formalize three logic structure types:

- **Parallel Structure.** A set of constraints $C = \{c_1, c_2, \ldots, c_n\}$ that must all be satisfied simultaneously. This structure corresponds to the flat assumption commonly adopted in prior work, where constraints are treated as independent(e.g., "Respond in English **and** use no commas **and** limit the length to 100 words").

- **Sequential Structure.** An ordered sequence of constraints $S = (c_1, c_2, \ldots, c_n)$, where each constraint c_t is meaningful only if all preceding constraints $(c_1, \ldots, c_{t-1})$ are successfully satisfied (e.g., "**First** generate an outline, **then** write a summary, **finally** translate it into English").

Logic Type	# Inst.	# Cons. Types	# Cons.	Evaluation
Parallel	17510	48	52106	
Sequential	10435	25	31295	
Conditional	10574	25	42152	

Table 1: Statistics of LSRINSTRUCT. #Inst., #Cons. Types, #Cons. and Evaluation refer to the number of instructions, constraint types, total constraints, and evaluation methods.

- **Conditional Structure.** A branching structure governed by a trigger constraint c_p. The active execution branch is determined by whether c_p is satisfied: if c_p holds, the model must satisfy the true-branch constraint c_{true}; else, it must satisfy the false-branch constraint c_{false} (e.g., "**If** the input text contains code, explain its functionality; **else**, summarize the text").

We construct the dataset by collecting seed instructions from Infinity-Instruct (Li et al., 2025), Open Assistant (Köpf et al., 2024), Self-Instruct (Wang et al., 2022a) and Super-Natural (Wang et al., 2022b), defining constraint types (hard constraints in Tab. 5, soft constraints in Tab. 6), and using GPT-4.1 to generate multi-constraint instructions that instantiate these logical structures. Each instruction follows logical structure with multiple constraints organized accordingly, enabling controlled analysis and structure-aware training. Detailed statistics of LSRINSTRUCT are shown in Tab. 1.

3.2 Structure-Aware Reward Modeling

We adopt the Group Relative Policy Optimization (GRPO) (Shao et al., 2024) training, where model optimization is driven by automatically computed signals indicating constraint satisfaction. For hard constraints, we use programmatic verification. For soft constraints, we employ a reward model to assess adherence. We train Qwen2.5-7B-Instruct as the reward model, where we exploit the natural partial order in and-type multi-constraint instructions to construct binary preference pairs and train the model via supervised fine-tuning with a binary classification objective following (Ren et al., 2025).

Given constraint-level verification results, we aggregate these rewards according to the logical structure of each instruction. Formally, let o denote a model output and c denote an atomic constraint. We define a binary verification function